

Orthoptic locus

locus of a point, the tangents from which to a conic include a right angle is called the orthoptic locus of the conic.

OR

The locus of the point of intersection of $\perp^r$ tangents to a conic is called orthoptic locus of the conic

Q. Find the locus of point of intersection of $\perp^r$ tangents to :

(a) parabola, $y^2 = 4ax$

(b) ellipse, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(c) hyperbola, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(a) let the point of intersection be (x_1, y_1) . Then the eqn to a parabola is $y = mx + \frac{a}{m}$. $\therefore$ it passes through (x_1, y_1)

$$\Rightarrow y_1 = mx_1 + \frac{a}{m}$$

$$\Rightarrow my_1 = m^2 x_1 + a$$

$$\Rightarrow m^2 x_1 - my_1 + a = 0 \quad \text{--- (1)}$$

eqn (1) is a quadratic in m . $\therefore$ it has two values of m , say m_1, m_2 which are the slopes of the tangents drawn from (x_1, y_1) to the parabola. $\therefore$ if the tangents are $\perp$.

$$\text{then } m_1 m_2 = -1$$

$$\Rightarrow \frac{a}{x_1} = -1$$

$$\Rightarrow \underline{x_1 + a = 0}$$

$\therefore$ locus of parabola is the line $\underline{x + a = 0}$ which is the directrix as well.

$$x^2 + 4x + 4 = 0$$

$$ax^2 + bx + c = 0$$

$$\text{product of roots} = \frac{c}{a}$$

$$\text{sum " " } = -\frac{b}{a}$$

(b) let the point of intersection be (x_1, y_1) . Then the eqn to an ellipse is $y = mx + \sqrt{a^2 m^2 + b^2}$. $\therefore$ it passes through (x_1, y_1)

$$\Rightarrow y_1 = mx_1 + \sqrt{a^2 m^2 + b^2}$$

$$\Rightarrow (y_1 - mx_1)^2 = a^2 m^2 + b^2$$

$$\Rightarrow m^2(x_1^2 - a^2) - 2mx_1 y_1 + y_1^2 - b^2 = 0$$

which is a quadratic in m . hence gives two values for m say m_1, m_2 which are the slopes of tangents passing through (x_1, y_1) to the ellipse. $\therefore$ the tangents are $\perp$

$$\Rightarrow m_1 m_2 = -1$$

$$\Rightarrow \frac{y_1^2 - b^2}{x_1^2 - a^2} = -1$$

$$\Rightarrow y_1^2 - b^2 = a^2 - x_1^2$$

$$y_1^2 - b^2 + x_1^2 - a^2 = 0$$

$$\underline{x_1^2 + y_1^2 = a^2 + b^2}$$

$x^2 + y^2 = a^2 + b^2$ is the orthoptic locus of an ellipse which is the circle and is known as director circle of the ellipse.

(c) Let the point of intersection be (x_1, y_1) . Then the eqn to a ~~hyperbola~~ parabola is $y = mx + \sqrt{a^2 m^2 - b^2}$. it passes through (x_1, y_1)

$$\Rightarrow (y_1 - mx_1)^2 = a^2 m^2 - b^2$$

$$\Rightarrow x_1^2 m^2 (x_1^2 - a^2) - 2mx_1 y_1 + y_1^2 + b^2 = 0$$

which is a quadratic eqn in m . hence it gives two values for m . Say m_1, m_2 which are the slopes of tangents passing through (x_1, y_1) to the parabola hyperbola. $\therefore$ the tangents are $\perp$

$$m_1 m_2 = -1$$

$$\Rightarrow \frac{y_1^2 + b^2}{x_1^2 - a^2} = -1$$

$$\Rightarrow y_1^2 + b^2 = a^2 - x_1^2$$

$$\Rightarrow \underline{x_1^2 + y_1^2 = a^2 - b^2}$$

$x^2 + y^2 = a^2 - b^2$ is the orthoptic locus of a hyperbola.

Q Find the eqn of the tangent at the point (x_1, y_1) on
 (a) parabola $y^2 = 4ax$ (b) ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (c) hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(a) Let (x_2, y_2) be any point on given parabola close to (x_1, y_1) . Then eqn of the line through (x_1, y_1) &

(x_2, y_2) is

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) \quad \text{--- (1)}$$

$\therefore (x_1, y_1)$ and (x_2, y_2) are points on parabola

$$y_1^2 = 4ax_1 \quad \text{--- (2)}$$

$$y_2^2 = 4ax_2 \quad \text{--- (3)}$$

$$(3) - (2) \Rightarrow (y_2 + y_1)(y_2 - y_1) = 4a(x_2 - x_1)$$

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{4a}{y_2 + y_1} \quad \text{--- (4)}$$

Sub (4) in (1) $\Rightarrow$

$$y - y_1 = \frac{4a}{y_1 + y_2} (x - x_1) \quad \text{--- (5)}$$

(making
 $x_1 = x_2$
 $y_1 = y_2$
by eliminating
them)

Now let the points (x_2, y_2) gradually approach and finally coincide with (x_1, y_1) . Then the chord joining (x_1, y_1) & (x_2, y_2) becomes the tangent at (x_1, y_1) .
Hence putting $y_1 = y_2$ --- (6) in (5)

$$\Rightarrow y - y_1 = \frac{4a}{2y_1} (x - x_1)$$

$$y_1(y - y_1) = 2a(x - x_1)$$

$$yy_1 - y_1^2 = 2ax - 2ax_1$$

$$yy_1 - 4ax_1 = 2ax - 2ax_1$$

$$yy_1 = 2a(x + x_1)$$

pt (0,3) on $y^2=3x$

$$x=0 \quad y=3$$

$$4a=3 \Rightarrow a=3/4$$

Sub in this eqn

This eqn of tangent at (x_1, y_1) .

b. Let (x_2, y_2) be any point on ellipse close to (x_1, y_1) . Then eqn of the line through them is

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) \quad \text{--- (1)}$$

$\therefore (x_1, y_1)$ and (x_2, y_2) are points on ellipse.

$$\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1 \quad \text{--- (2)}$$

$$\frac{x_2^2}{a^2} + \frac{y_2^2}{b^2} = 1 \quad \text{--- (3)}$$

$$(3) - (2) \Rightarrow \frac{x_2^2 - x_1^2}{a^2} + \frac{y_2^2 - y_1^2}{b^2} = 0$$

$$\Rightarrow \frac{(x_2 - x_1)(x_2 + x_1)}{a^2} + \frac{(y_2 - y_1)(y_2 + y_1)}{b^2} = 0$$

$$\Rightarrow \frac{-b^2(x_2 + x_1)(y_2 + y_1)}{a^2} = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{--- (4)}$$

Sub (4) in (1)

$$\Rightarrow y - y_1 = \frac{-b^2}{a^2} \left(\frac{x_2 + x_1}{y_2 + y_1} \right) (x - x_1)$$

Now let the points coincide each other gradually
 $\Rightarrow x_1 = x_2$ & $y_1 = y_2$. Now the chord is the tangent.

$$\Rightarrow y - y_1 = \frac{-b^2}{a^2} \frac{x_1}{y_1} (x - x_1)$$

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$$\Rightarrow yy_1 - y_1^2 = \frac{-b^2}{a^2} (xx_1 - x_1^2)$$

$$\Rightarrow \frac{yy_1}{b^2} - \frac{y_1^2}{b^2} = \frac{x_1^2}{a^2} - \frac{xx_1}{a^2}$$

$$\Rightarrow \frac{yy_1}{b^2} + \frac{xx_1}{a^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$$

$$\Rightarrow \frac{yy_1}{b^2} + \frac{xx_1}{a^2} = 1$$

(10) Let (x_2, y_2) be any point on hyperbola close to (x_1, y_1) . Then eqn through them is

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) \quad \text{--- (1)}$$

$\therefore (x_1, y_1)$ and (x_2, y_2) lies on hyperbola.

$$\Rightarrow \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1 \quad \text{--- (2)}$$

$$\Rightarrow \frac{x_2^2}{a^2} - \frac{y_2^2}{b^2} = 1 \quad \text{--- (3)}$$

(3) - (2)

$$\Rightarrow \frac{x_2^2 - x_1^2}{a^2} - \frac{y_2^2 - y_1^2}{b^2} = 0$$

$$\frac{x_2^2}{a^2} - \frac{y_2^2}{b^2} - \left[\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} \right] = 0$$

$$\frac{x_2^2 - x_1^2}{a^2} + \frac{y_1^2 - y_2^2}{b^2} = 0$$

$$\frac{(x_2 - x_1)(x_2 + x_1)}{a^2} + \frac{(y_1 + y_2)(y_1 - y_2)}{b^2} = 0$$

$$\frac{(x_2 - x_1)}{a^2} \cdot (x_2 - x_1)(x_2 + x_1) = - \frac{(y_1 + y_2)(y_1 - y_2)}{b^2}$$

$$\frac{b^2(x_1 + x_2)}{a^2(y_1 + y_2)} = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{--- (1)}$$

$$\text{Sub (1) in (1)} \Rightarrow$$

$$\frac{y - y_1}{x - x_1} = \frac{b^2(x_1 + x_2)}{a^2(y_1 + y_2)}$$

Now let the points coincide gradually making the chord the tangent $\Rightarrow y_1 = y_2$ and $x_1 = x_2$.

$$\Rightarrow \frac{y - y_1}{x - x_1} = \frac{b^2 x_1}{a^2 y_1}$$

$$\Rightarrow \frac{y y_1}{b^2} - \frac{y_1^2}{b^2} = \frac{x^2}{a^2} - \frac{x_1^2}{a^2}$$

$$\Rightarrow \frac{-y y_1}{b^2} + \frac{x x_1}{a^2} = 1$$

$$\frac{x_2^2}{a^2} - \frac{y_2^2}{b^2} - \left[\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} \right] = 0$$

$$\frac{x_2^2 - x_1^2}{a^2} + \frac{y_1^2 - y_2^2}{b^2} = 0$$

$$\frac{(x_2 - x_1)(x_2 + x_1)}{a^2} + \frac{(y_1 + y_2)(y_1 - y_2)}{b^2} = 0$$

$$\cancel{(x_2 - x_1)} \cdot \frac{(x_2 + x_1)}{a^2} = - \frac{(y_1 + y_2)(y_1 - y_2)}{b^2}$$

$$\frac{b^2(x_1 + x_2)}{a^2(y_1 + y_2)} = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{--- (4)}$$

Sub(4) in (1) $\Rightarrow \frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$

$$\frac{y - y_1}{x - x_1} = \frac{b^2(x_1 + x_2)}{a^2(y_1 + y_2)}$$

Now let the points coincide gradually making the chord the tangent $\Rightarrow y_1 = y_2$ and $x_1 = x_2$.

$$\Rightarrow \frac{y - y_1}{x - x_1} = \frac{b^2 x_1}{a^2 y_1}$$

$$\Rightarrow \frac{y y_1}{b^2} - \frac{y_1^2}{b^2} = \frac{x x_1}{a^2} - \frac{x_1^2}{a^2}$$

$$\Rightarrow \frac{-y y_1}{b^2} + \frac{x x_1}{a^2} = 1$$